



Wind Turbines in the U.S.

Data Analysis & Visualization Capstone Project

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Goals of the Project

In this project I analyzed U.S. wind turbine data to identify trends in turbine capacity, technology development, and geographic distribution over time, using interactive visualizations to present key insights.



Understanding the problem

Why Wind Energy?

Wind energy is one of the fastest-growing sources of renewable power in the world.

It helps reduce greenhouse gas emissions, decreases reliance on fossil fuels, and supports energy sustainability goals.

In the U.S., thousands of wind turbines are already installed across many states, but their distribution and capacity vary widely

Why This Analysis Matters:

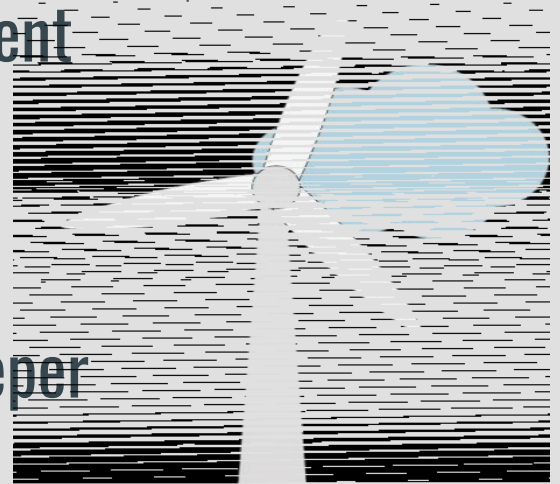
- Identifies which states are leading in wind energy development
- Highlights potential for growth in underrepresented regions
- Supports data-driven decision making for policy makers, investors, and engineers
- Encourages public awareness by visualizing clean energy progress

Project Impact:

By analyzing and visualizing this data, we can better understand how wind power is shaping the U.S. energy landscape — and where it can grow next.

Project objective:

- Explore trends in wind turbine deployment across U.S. states
- Analyze total turbine capacity and geographic distribution
- Create interactive visualizations for deeper insights



Dataset Overview

```
[ ] 1 import pandas as pd
    2 df = pd.read_csv('wind_turbines.csv')
    3 df.head()
```

	Site.State	Site.County	Year	Turbine.Capacity	Turbine.Hub_Height	Turbine.Rotor_Diameter	Turbine.Swept_Area	Turbine.Total_Height	Project.Capacity	Project.Number_Turbines	Site.Latitude	Site.Longitude
0	IA	Story County	2017	3000	87.5	125.0	12271.85	150.0	30.0	10	-93.518082	42.013630
1	IA	Hardin County	2017	3000	87.5	125.0	12271.85	150.0	30.0	10	-93.367798	42.497940
2	IA	Story County	2017	3000	87.5	125.0	12271.85	150.0	30.0	10	-93.513710	42.019119
3	IA	Story County	2017	3000	87.5	125.0	12271.85	150.0	30.0	10	-93.523651	42.006813
4	IA	Story County	2017	3000	87.5	125.0	12271.85	150.0	30.0	10	-93.632835	41.882477

This dataset includes over 63,000 entries, each representing an individual wind turbine installed in the U.S. It contains details such as:

- Location — State, County, Latitude, and Longitude
- Technical Specifications — Hub Height, Rotor Diameter, Swept Area, Total Height
- Performance — Individual Turbine Capacity and Total Project Capacity
- Installation Info — Year of commissioning and number of turbines in the project

Data Source: [CORGIS Wind Turbines Dataset](#)

Tools Used

Tech Stack:

- Python (Pandas, Bokeh, Plotly)
- Google Colab and Jupyter Notebook for analysis
- GitHub for version control
- Netlify to publish visuals
- PowerPoint to present insights

Data cleaning and transformation

- The dataset was clean — no missing or duplicate values were found.
- Renamed several columns to improve clarity and readability (e.g., 'Site.State': 'State', 'Site.County': 'County').
- Ensured date formats were consistent for analysis.
- Applied filtering later during visualization to focus on specific states, years, and capacity ranges.

```
▼ Check for Duplicates

[ ] 1 print(df.duplicated().sum())

0

We don't have any duplicates.

▼ Rename Columns for Clarity

[ ] 1 df.rename(columns={
2     'Site.State': 'State',
3     'Site.County': 'County',
4     'Turbine.Capacity': 'Turbine_Capacity_MW',
5     'Turbine.Hub_Height': 'Hub_Height_m',
6     'Turbine.Rotor_Diameter': 'Rotor_Diameter_m',
7     'Turbine.Swept_Area': 'Swept_Area_m2',
8     'Turbine.Total_Height': 'Total_Height_m',
9     'Project.Capacity': 'Project_Capacity_MW',
10    'Project.Number_Turbines': 'Number_of_Turbines',
11    'Site.Latitude': 'Latitude',
12    'Site.Longitude': 'Longitude'
13 }, inplace=True)
14 df
```

	State	County	Year	Turbine_Capacity_MW	Hub_Height_m	Rotor_Diameter_m	Swept_Area_m2	Total_Height_m	Project_Capacity_MW	Number_of_Turbines	Latitude	Longitude
0	IA	Story County	2017	3000	87.5	125.0	12271.85	150.0	30.000	10	-93.518082	42.013630
1	IA	Hardin County	2017	3000	87.5	125.0	12271.85	150.0	30.000	10	-93.367798	42.497940
2	IA	Story County	2017	3000	87.5	125.0	12271.85	150.0	30.000	10	-93.513710	42.019119
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Trend analysis

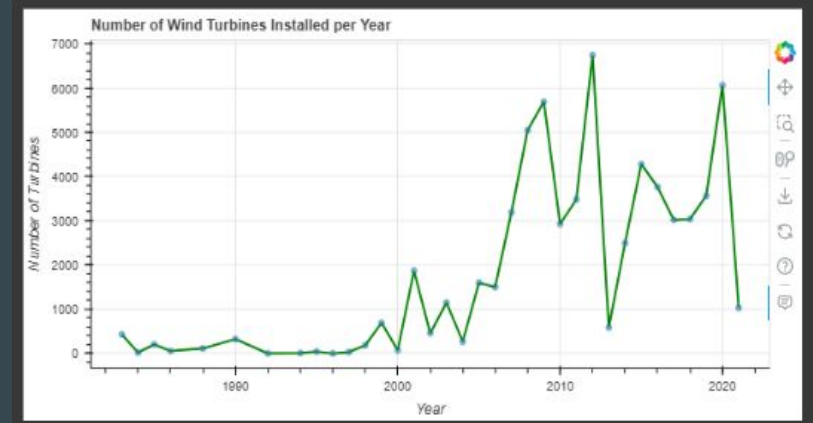
→ Let's see how wind turbine installations have changed over

```
1 # Group the dataset by year and count the number of entries (turbines) per year
2 turbines_per_year = df.groupby("Year").size().reset_index(name="Number_of_Turbines")
3 turbines_per_year.head()
```

	Year	Number_of_Turbines
0	1983	432
1	1984	20
2	1985	205
3	1986	56
4	1988	115

дальші дії: [Створити код зі змінною turbines_per_year](#) [Переглянути рекомендовані графіки](#) [New interactive sheet](#)

```
1 from bokeh.plotting import figure, show, output_notebook
2 from bokeh.models import ColumnDataSource
3
4 output_notebook()
5
6 # Wrap the data in a ColumnDataSource so Bokeh can use it
7 source = ColumnDataSource(turbines_per_year)
8
9 p = figure(title="Number of Wind Turbines Installed per Year",
10           x_axis_label='Year',
11           y_axis_label='Number of Turbines',
12           height=350, width=700, tooltips=[('Year', '@Year'), ('Number of Turbines', '@Number_of_Turbines')])
13
14 p.line(x='Year', y='Number_of_Turbines', source=source, line_width=2, color="green")
15 p.scatter(x='Year', y='Number_of_Turbines', source=source, size=5, alpha=0.5)
16
17 show(p)
```

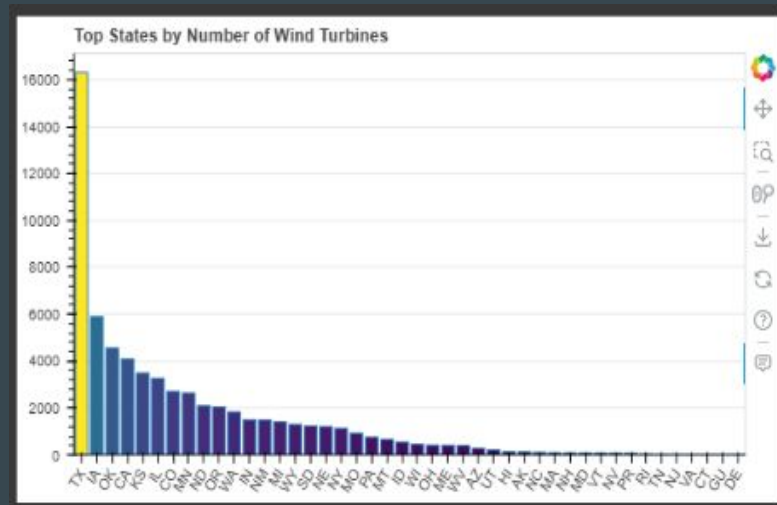


Turbine Distribution by State

- Texas leads in wind energy — likely because of its large size, windy areas, and open land.
- Wind power is concentrated in a few states with good natural conditions.
- There's a sharp drop after the top states, meaning wind energy is not spread evenly across the country.



```
5
6 # counts how many wind turbines are present in each state using the State column.
7 state_counts = df['State'].value_counts()
8 #Converts the index (state abbreviations) into a list of strings.
9 state_names = state_counts.index.astype(str).tolist()
10
11 #Creating a dictionary with two keys:
12 data = {
13     'states': state_names,
14     'counts': state_counts.values
15 }
16
17 source = ColumnDataSource(data)
18
19 # Create the color map using Viridis
20 color_map = linear_cmap(field_name='counts', palette=Viridis256, low=min(state_counts.values), high=max(state_counts.values))
21
22 # Create figure
23 p2 = figure(x_range=state_names, height=350, title="Top States by Number of Wind Turbines",
24             tooltips=[('State', '@states'), ('Number of Turbines', '@counts')])
25
26 # Create the bar chart with continuous color mapping using Viridis
27 p2.vbar(x='states', top='counts', width=0.8, source=source, fill_color=color_map)
28
29 # Customize the chart
30 p2.xgrid.grid_line_color = None
31 p2.y_range.start = 0
32 p2.xaxis.major_label_orientation = 1
33
34 # Show the plot
35 show(p2)
```

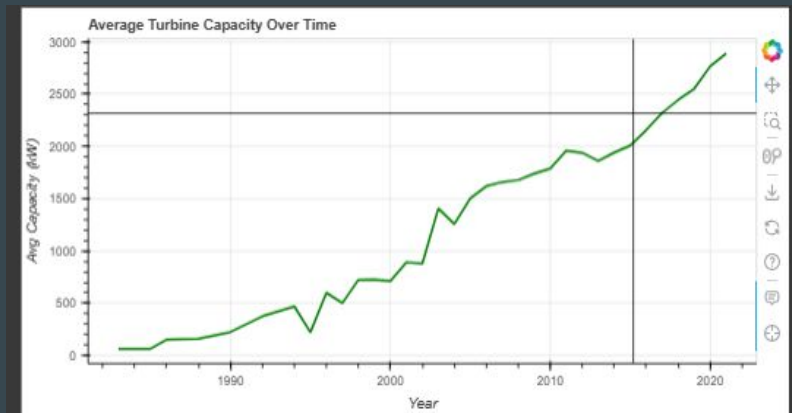


Analyze average turbine capacity by year.

This chart shows how the average capacity (in kilowatts) of wind turbines installed each year has changed over time. What it tells us:

- The capacity of turbines has steadily increased since the 1980s.
- Especially after the early 2000s, we see a sharp rise, which likely reflects improvements in turbine technology and a push for more efficient energy generation.
- This suggests that rather than just building more turbines, the U.S. has also focused on building better turbines.
- In short: Turbines are getting stronger and smarter over time.

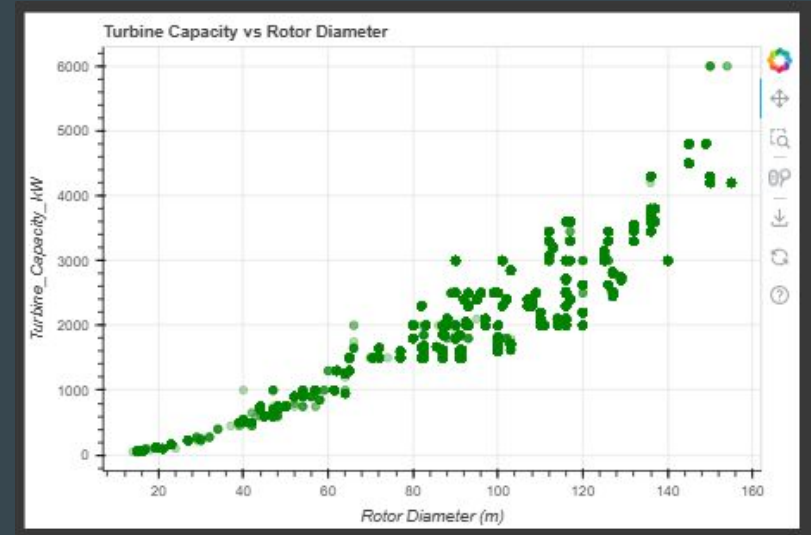
```
1 from bokeh.models import ColumnDataSource, CrosshairTool
2
3 # Prepare the data
4 avg_capacity_by_year = df.groupby('Year')['Turbine_Capacity_kW'].mean().reset_index()
5 source = ColumnDataSource(avg_capacity_by_year)
6
7 # Create the plot
8 p3 = figure(title="Average Turbine Capacity Over Time",
9             x_axis_label='Year',
10             y_axis_label='Avg Capacity (kW)',
11             height=350, width=700,
12             tooltips=[('Year', '@Year'), ('Avg Capacity (kW)', '@Turbine_Capacity_kW{0.0}')])
13
14 # Line plot
15 p3.line(x='Year', y='Turbine_Capacity_kW', source=source, line_width=2, color="green")
16
17 # Add crosshair tool
18 p3.add_tools(CrosshairTool())
19
20 # Show plot
21 show(p3)
```



Turbine Capacity vs Rotor Diameter

- There's a clear positive correlation — newer turbines with larger rotors are designed to be more powerful. This shows how engineering advancements are making turbines more efficient over time.

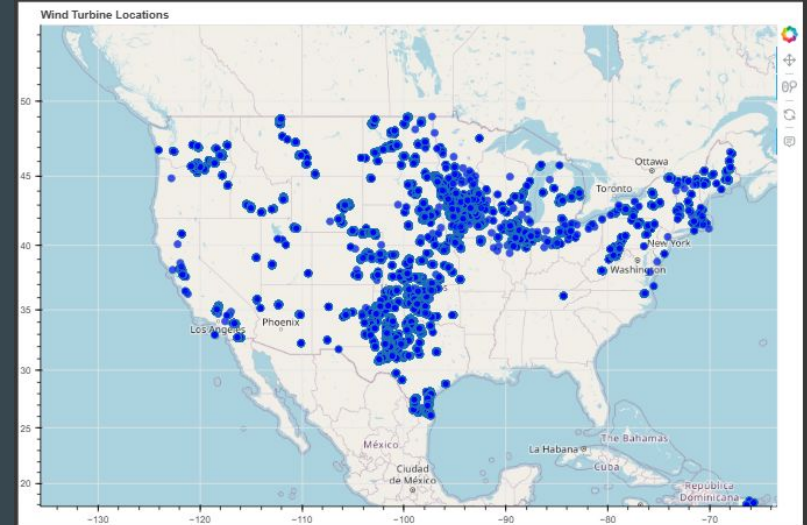
```
1 p4 = figure(title="Turbine Capacity vs Rotor Diameter", x_axis_label='Rotor Diameter (m)',  
2           y_axis_label='Turbine Capacity_kW',  
3           height=400, width=600)  
4 p4.scatter(df['Rotor_Diameter_m'], df['Turbine_Capacity_kW'], size=6, color="green", alpha=0.3)  
5  
6 show(p4)
```



Wind Turbine locations across the US.(Point Map)

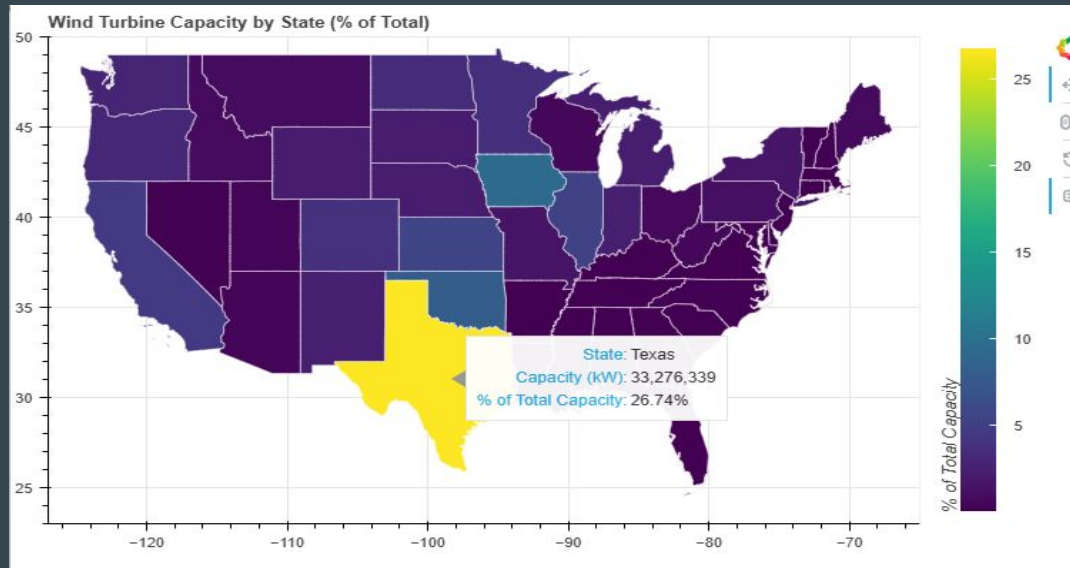
- The map visualizes wind turbine locations across the US.
- The majority of wind turbine installations are concentrated in the central and western parts of the United States, particularly in states like Texas, Oklahoma, Iowa, and Kansas, which are known for their favorable wind conditions and flat terrains.
- The eastern seaboard and Great Lakes region also show notable activity.
- A few turbines are located in Alaska, Hawaii, and U.S. territories, indicating offshore or island-based turbines.

```
1 from bokeh.io import output_file, show
2 from bokeh.models import ColumnDataSource, HoverTool, WMTSTileSource
3 from bokeh.plotting import figure
4 from pyproj import Transformer
5
6 # Convert lat/lon to Web Mercator - This converts geographic coordinates (Latitude, Longitude)
7 # into Web Mercator (x, y), which is required for plotting on tile maps like OpenStreetMap.
8 transformer = Transformer.from_crs("epsg:4326", "epsg:3857")
9 # EPSG:4326 is standard lat/lon, EPSG:3857 is Web Mercator projection.
10 df["x"], df["y"] = transformer.transform(df["Longitude"].values, df["Latitude"].values)
11
12 source = ColumnDataSource(df)
13
14 # Set up tile source manually Sets the map layer (tiles) from OpenStreetMap.
15 # These {Z}/{X}/{Y} values dynamically load tiles depending on zoom and location.
16
17 #Bokeh needs a tile provider to display geographical context
18 tile_url = "http://c.tile.openstreetmap.org/{Z}/{X}/{Y}.png"
19 tile_source = WMTSTileSource(url=tile_url)
20
21 # Set up plot
22 p5 = figure(title="Wind Turbine Locations", x_axis_type="mercator", y_axis_type="mercator",
23            width=900, height=600, tools="pan,wheel_zoom,reset")
24
25 p5.add_tile(tile_source)
26 p5.scatter(x="x", y="y", size=8, fill_color="blue", fill_alpha=0.7, source=source)
27
28 hover = HoverTool()
29 hover.tooltips = [
30     ("State", "@State"),
31     ("Year", "@Year"),
32     ("Capacity (kW)", "@Turbine_Capacity_kW"),
33 ]
34 p5.add_tools(hover)
35
36 output_file("wind_turbine_point_map.html")
37 show(p5)
```



Wind Turbine Capacity by State (% of Total) Map

- This interactive choropleth map visualizes each U.S. state's contribution to total wind turbine capacity.
- ♦ Texas leads with 33 million kW, making up nearly 27% of the total U.S. capacity.
- ♦ Brighter colors indicate higher contributions, while darker shades show lower capacity.
- This view helps identify geographic patterns and leadership in wind energy adoption.



[Interactive Map of Wind Turbine Capacity by State \(% of Total\)](#)

Insights and Key Takeaways

Deliverable 1

Texas is the undisputed leader in wind energy, contributing over 26% of the total U.S. turbine capacity — more than double that of any other state.

Deliverable 2

Wind energy development is highly concentrated in the Midwest and Central U.S., especially in states like Iowa, Oklahoma, and Kansas, which benefit from favorable wind conditions and land availability.

Deliverable 3

Southern and Southeastern states show minimal wind energy adoption, highlighting potential gaps in infrastructure, policy, or geography that could be targeted for development.

Deliverable 4

The number of wind turbines installed annually peaked in the 2010s, showing a rapid expansion phase — a key trend for forecasting future growth or policy impact.

Business Recommendations:



- Invest in high-capacity states (e.g., Texas, Iowa) to leverage existing infrastructure and workforce readiness.
- Explore underserved states (e.g., Alabama, Louisiana) to pioneer wind energy services where competition is low.
- Develop location-based service models using turbine data (maintenance, analytics, drone inspection).
- Collaborate with policymakers in emerging regions to shape incentives and regulatory frameworks.
- Use data storytelling (like this project) to pitch renewable strategies to stakeholders and clients.

Thank You!

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Questions? I'm happy to discuss the project.